

Error Analysis 🤖

After bias/variance analysis, **error analysis** is the second most important diagnostic for deciding what to work on next. It's a manual process of examining the mistakes your algorithm is making to find systematic patterns.

The Error Analysis Process

1. Get a Sample of Misclassified Examples:

- Start with your cross-validation (CV) set.
- Let's say your model misclassifies 100 out of 500 CV examples.
- If you have too many errors (e.g., 1,000), just take a random sample (e.g., 100) to analyze.

2. Manually Examine and Categorize the Errors:

- Look through each of the 100 mistakes and try to "tag" them with a hypothesis for *why* the error occurred.
- **Example (Spam Classification):**
 - **Type of Error:** Pharmaceutical (pharma) spam
 - **Count:** 21 / 100
- **Type of Error:** Password stealing (phishing) emails
 - **Count:** 18 / 100
- **Type of Error:** Has unusual email routing
 - **Count:** 7 / 100
- **Type of Error:** Spam message hidden in an image
 - **Count:** 5 / 100
- **Type of Error:** Deliberate misspellings (e.g., "w4tches")
 - **Count:** 3 / 100
- **Note:** These categories can overlap (e.g., a pharma email might also have misspellings).

3. Use the Counts to Prioritize Your Work:

- The counts immediately show you where the biggest problems are.
- **High-Impact:** Pharma spam (21%) and phishing emails (18%) are the largest sources of error. Working on these will give you the biggest improvement.
- **Low-Impact:** Deliberate misspellings only account for 3% of the errors. Even if you

build a perfect misspelling detector, it will only fix 3 out of your 100 errors. This should be a **low priority**.

How Error Analysis Guides Your Next Steps

Error analysis gives you concrete inspiration for what to do.

- **If "Pharma Spam" is the top error:**
 - **New Feature Idea:** Create a feature that detects specific drug or pharmaceutical product names.
 - **New Data Idea:** Go out and *specifically* collect more data of pharma spam emails to help your model learn to recognize them.
- **If "Phishing" is the top error:**
 - **New Feature Idea:** Create a feature that analyzes URLs in the email to see if they are suspicious.
 - **New Data Idea:** Collect more examples of phishing emails.

This process is far more efficient than just "collecting more data" in general, as it allows you to target your efforts on the highest-impact problems.

Limitation: Error analysis is much easier to perform on tasks that humans are also good at (like spam detection, vision, or speech) because you can use your own intuition to spot the patterns. It's much harder for problems humans can't easily solve (like predicting ad clicks).